Classifier Challenges

Construct an attack. Check the evidence.

Artificial Intelligence III / 15 October 2026

These exercises accompany the written course notes on attack mechanisms. Work in pairs on the classifier assigned to you.

How to work

Spend the first three minutes predicting the result individually. Then compare your predictions. One partner proposes the perturbation or argument; the other checks the label, input domain, and total perturbation budget. Switch roles halfway through. Finish by exchanging your one-page result with another pair.

Classifier 1 / The same radius under two norms

Classifier. Let $\mathcal{X} = [0,1]^2$ and

$$g(\mathbf{x}) = x_1 + 2x_2 - 1, \qquad \mathbf{x}_0 = (0.60, 0.40)^\top.$$

Predict C_1 when $g(\mathbf{x}) \geq 0$ and C_2 when $g(\mathbf{x}) < 0$. The attacker seeks a strict change from C_1 to C_2.

1. **Predict and draw.** Compute the clean score and label. Sketch the boundary, mark $\mathbf{x}_0$, and indicate the direction that decreases g fastest under an ℓ_2 constraint. Before using any tools, predict whether radius $\eta = 0.15$ is enough to cross.

2. **Construct an attack.** Under $\|\mathbf{r}\|_\infty \leq 0.15$ and $\mathbf{x}_0 + \mathbf{r} \in \mathcal{X}$, find a valid attack. Report $\mathbf{r}$, the attacked input, both its ℓ_∞ and ℓ_2 norms, and the final score and label.

3. **Change the norm.** Replace the constraint by $\|\mathbf{r}\|_2 \leq 0.15$. Is any successful attack possible? Prove your answer with a bound or an exact boundary distance; a failed search is insufficient.

4. **Find the thresholds.** Compute the minimum distance to the boundary in each norm. Give a boundary-reaching perturbation for each. Explain whether these distances themselves change the label under the stated tie convention. Add the two perturbation regions to your sketch.

5. **Audit the claim.** Evaluate: "Equal numerical ℓ_2 and ℓ_∞ budgets allow the same attacks." Give a counterexample or proof.